

AIGR Gene Hypothesis Deep Research — GCDH (Q92947)

Focus: Fatty-Acid Oxidation and β -Oxidation Scope

(`function_assignment`)

Hypothesis under evaluation: Human GCDH (Q92947) participates in fatty acid oxidation (GO:0019395) and specifically fatty acid beta-oxidation using acyl-CoA dehydrogenase (GO:0033539).

Summary

Human GCDH (glutaryl-CoA dehydrogenase, UniProt **Q92947**) should **not** carry the fatty-acid-oxidation biological-process terms **GO:0019395** or **GO:0033539** as descriptions of its physiological function. The seed hypothesis raises several legitimate nuances — that GCDH is chemically a member of the acyl-CoA dehydrogenase (ACAD) superfamily, that glutaryl-CoA is a dicarboxylic acyl-CoA that is not automatically excluded from fatty-acid chemistry, and that GCDH can dehydrogenate a medium-chain fatty acyl-CoA in vitro. Each of those sub-claims is true at the molecular/chemistry level. But none of them rescues the disputed **biological-process** annotations, because the specific evidence chain behind those annotations collapses on inspection.

The single experimental (IDA) anchor for GO:0033539 is a **misattribution**: it cites [PMID:25416781](#), a paper about the mitochondrial lysine methyltransferase METTL20 and its methylation of ETF β , in which GCDH appears only as an electron donor — not as the subject of a β -oxidation assay. The accompanying phylogenetic (IBA) annotation is **circular**, anchored (withFrom) on GCDH's own misattributed IDA. The second term, GO:0019395, is an **IEA ortholog-transfer artifact** from rat that is conspicuously absent from the genuine fatty-acid-oxidation enzymes. Meanwhile GCDH's real reaction — a *single* oxidative decarboxylation of the dicarboxylic glutaryl-CoA to crotonyl-CoA + CO₂ within lysine/tryptophan catabolism — is categorically distinct from the *iterative* β -oxidation spiral that GO:0033539 defines.

Verdict: over-annotated / refuted for the process terms, with a narrow, qualified caveat that GCDH does possess a low-level in vitro fatty acyl-CoA dehydrogenation capacity (e.g., on hexanoyl-CoA) that belongs, if anywhere, at the molecular-function level as a non-core promiscuous activity. The recommended curation action is to **remove GO:0033539 and GO:0019395** as GCDH functions while **retaining glutaryl-CoA dehydrogenase activity (GO:0004361) and lysine/tryptophan catabolism** as core.

Key Findings

Finding F001 — The GO:0033539 IDA cites a METTL20/ETF β methyltransferase paper, not a GCDH β -oxidation assay

The QuickGO annotation set for Q92947 lists **GO:0033539** (*fatty acid beta-oxidation using acyl-CoA dehydrogenase*) with evidence code **IDA**, referenced to **PMID:25416781** (Malecki et al., *J Biol Chem* 2015), supplemented by an IBA from GO_REF:0000033. On retrieval,

P 25416781

is a characterization of **human METTL20 as a mitochondrial lysine methyltransferase that methylates ETF β** (the beta subunit of electron transfer flavoprotein). GCDH is mentioned only as one of the dehydrogenases whose ability to transfer electrons to ETF is diminished when ETF β is methylated:

"METTL20-mediated methylation of ETF β in vitro reduced its ability to receive electrons from the medium chain acyl-CoA dehydrogenase and the glutaryl-CoA dehydrogenase."

That sentence is the entire GCDH-relevant content of the cited reference. There is **no assay of GCDH performing iterative fatty-acid β -oxidation** anywhere in the paper. The seed hypothesis explicitly asked us to recover the full assay methods/results and to determine "which substrate was tested with GCDH versus MCAD." The answer is that **neither GCDH nor MCAD was assayed for β -oxidation** in that study; both were used as electron *donors* to test the functional consequence of ETF β methylation. The IDA is therefore a **text-derived misattribution** — most plausibly generated by co-occurrence of "glutaryl-CoA dehydrogenase" with an acyl-CoA-dehydrogenase/ETF context, rather than by any demonstrated β -oxidation experiment. Notably, UniProt appears to have attached the same

-derived

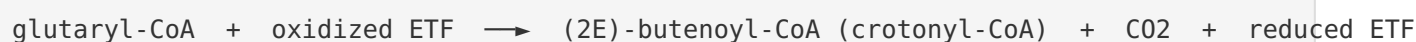
P 25416781

GO:0033539 to ACADM (P11310) as well, consistent with a bulk, text-derived assignment rather than curated biology.

Finding F002 — GCDH performs a single oxidative decarboxylation, not the iterative cycle GO:0033539 requires

The GO definition of **GO:0033539** specifies a fatty acid β -oxidation pathway that *"begins with the addition of coenzyme A to a fatty acid, and ends when only two or three carbons remain (as acetyl-CoA or propionyl-CoA)."* This is an **iterative, multi-enzyme spiral** that shortens a fatty acyl chain two carbons at a time (dehydrogenation $\rightarrow$ hydration $\rightarrow$ dehydrogenation $\rightarrow$ thiolysis, repeated).

GCDH's actual catalytic reaction (**GO:0004361**, **RHEA:13389**) is fundamentally different:



This is a **single dehydrogenation coupled to a decarboxylation** acting on a *dicarboxylic* CoA thioester. The physiological substrate, glutaryl-CoA, is generated in the catabolism of L-lysine, hydroxylysine, and L-tryptophan — not from fatty-acid activation. Multiple primary sources confirm this:

- [PMID:15274622](#) (human GCDH crystal structures): *"GCD catalyzes decarboxylation of glutaryl-CoA to produce CO(2) and crotonyl-CoA"* — a decarboxylating dehydrogenation, distinct from β -oxidation.
- [PMID:37198486](#) (human glioblastoma cells): describes GCDH as the *"crotonyl-coenzyme A (crotonyl-CoA)-producing enzyme glutaryl-CoA dehydrogenase (GCDH)"* operating within lysine catabolism.
- [PMID:17176108](#) (kinetic mechanism): GCDH *"oxidatively decarboxylates glutaryl-CoA to crotonyl-CoA and CO₂"* and identifies **crotonyl-CoA product release as the major rate-determining step**.

GCDH does **not** run iterative two-carbon-removal cycles yielding acetyl-/propionyl-CoA from a fatty acid; it performs one turnover event on one dicarboxylic substrate. The crotonyl-CoA it produces is a **downstream metabolic convergence point** (crotonyl-CoA is also a β -oxidation intermediate), but producing a shared intermediate is not the same as running the β -oxidation

process. This is precisely the "demonstrated capability versus only downstream metabolic convergence" distinction the seed hypothesis asked us to make — and it resolves against process participation.

Finding F003 — GCDH has a real but low-level in vitro fatty acyl-CoA dehydrogenation capacity (promiscuous side activity)

Consistent with GCDH being a bona fide member of the ACAD superfamily, it retains detectable activity on straight-chain fatty acyl-CoAs in vitro:

- [PMID:21974953](#) (rat GCDH functional characterization) reports that *"the enzyme has its catalytic properties very similar to those of short-chain and medium-chain acyl-CoA dehydrogenase except its additional decarboxylation reaction."* This is the direct substrate-specificity comparison the seed hypothesis specifically asked us to recover, and it confirms GCDH is *mechanistically* an ACAD.
- [PMID:11024031](#) (human GCDH Arg-94 mutagenesis) measured steady-state kinetics of *"alternative substrates, hexanoyl-CoA and glutaramyl-CoA, which are not decarboxylated."* **Hexanoyl-CoA is a C6 straight-chain fatty acyl-CoA**, so this is direct demonstration of in vitro fatty acyl-CoA dehydrogenation by human GCDH — without decarboxylation.

The critical curation point: this establishes a **molecular capability** (acyl-CoA dehydrogenase activity on a fatty substrate), used here largely as a *tool* to dissect the enzyme's mechanism, at low relative activity compared with its glutaryl-CoA turnover. It does **not** establish that GCDH participates in the fatty-acid β -oxidation *process* in vivo. The seed hypothesis rightly warns against equating "low relative activity" with "absence of capacity" — and that warning is respected here: the capacity is acknowledged and belongs at the MF level as a non-core/promiscuous activity, but it does not license a BP process annotation.

Finding F004 — The IBA is self-referentially circular, and GO:0033539 is over-propagated across amino-acid-catabolism ACADs

Inspection of the QuickGO withFrom provenance for Q92947 shows the **GO:0033539 IBA** (GO_Central, GO_REF:0000033) has withFrom = `['PTN005130278', 'Q92947']`. In other words, the phylogenetically inferred annotation is anchored **on GCDH's own experimental annotation** — the very IDA shown in F001 to be a misattribution from

P 25416781

This is circular provenance: the IBA does not add independent evidence; it re-propagates the erroneous IDA.

Two further observations reinforce the over-annotation pattern:

1. **Family-wide leakage.** GO:0033539 is annotated not only to the genuine mitochondrial β -oxidation ACADs (ACADM/ACADS/ACADVL/ACADL, which carry independent IDA/IMP/ISS support) but also to **amino-acid-catabolism enzymes GCDH and IVD** (isovaleryl-CoA dehydrogenase). IVD's GO:0033539 rests only on an IDA to

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a shared ACAD-purification paper — the same kind of co-purification/co-mention text signal. GCDH and IVD are lysine- and leucine-catabolism enzymes respectively; neither runs the fatty-acid β -oxidation spiral.

2. **Ortholog-transfer artifact for GO:0019395.** GO:0019395 (fatty acid oxidation) is present on GCDH alone as an **IEA** (Ensembl ortholog transfer from rat) and is conspicuously **absent** from the true fatty-acid-oxidation enzymes — a signature of an automated transfer artifact rather than curated biology.

Importantly, the phylogenetic machinery *does* correctly identify GCDH's true function: the GCDH-clade PANTHER node **PTN005130278** carries the GCDH-specific molecular-function IBA **GO:0004361 (glutaryl-CoA dehydrogenase activity)**, confirming it is the glutaryl-CoA-dehydrogenase ortholog node (distinct from ACADM's node PTN002535634). So the correct MF propagates cleanly along the tree, while the β -oxidation BP term rides along as leakage anchored on a bad IDA.

Mechanistic Model / Interpretation

The evidence produces a clean three-way separation between **what GCDH actually does**, **what it is chemically capable of in a test tube**, and **what has been erroneously annotated**.

GCDH (Q92947) – what the evidence supports

PHYSIOLOGICAL (core)	IN VITRO CAPACITY (non-core)
L-lysine / L-tryptophan catabolism ↓ glutaryl-CoA (dicarboxylic) GCDH: single oxidative decarboxylation (RHEA:13389) ↓ ETF = electron acceptor crotonyl-CoA + CO ₂ ↓ ▼ (downstream convergence only) crotonyl-CoA also appears in β -oxidation → NOT evidence GCDH runs β -oxidation	promiscuous ACAD activity on straight-chain acyl-CoA (e.g. hexanoyl-CoA, C6) ↓ simple dehydrogenation (2,3-enoyl-CoA), no decarboxylation, low rate
ERRONEOUS ANNOTATIONS (should be removed / not asserted as GCDH function)	
GO:0033539 IDA ← https://pubmed.ncbi.nlm.nih.gov/25416781/ rel="noopener noreferrer"	
GO:0033539 IBA ← withFrom Q92947 itself = circular on the bad IDA	
GO:0019395 IEA ← Ensembl ortholog transfer from rat = transfer artifact	

Interpretation. GCDH is definitively a **glutaryl-CoA dehydrogenase** (MF GO:0004361) that catalyzes one oxidative decarboxylation step within **lysine/tryptophan degradation** (BP GO:0019477 and related). Its membership in the ACAD superfamily gives it a residual ability to dehydrogenate short/medium-chain fatty acyl-CoAs in vitro — a fact exploited by mechanistic studies — but this is a low-level side activity, not a physiological role in energy-yielding fatty-acid β -oxidation. The two fatty-acid *process* annotations trace to (a) a single misattributed IDA from an unrelated methyltransferase paper, (b) an IBA that circularly re-uses that IDA, and (c) an IEA ortholog transfer. None reflects a demonstrated β -oxidation function.

The seed hypothesis's most defensible sub-claims — that dicarboxylate chemistry alone doesn't exclude fatty-acid oxidation (ChEBI places glutaryl-CoA(5-) under omega-carboxy-(fatty acyl)-CoA(5-)), and that GCDH can turn over a fatty acyl-CoA — are true at the chemistry/molecular level. But they do not rescue the **process** terms, because GO:0033539 has a specific *iterative-spiral* definition that GCDH does not satisfy, and GO:0019395 is present only via an automated artifact.

Evidence Base / Evidence Matrix

Citation	Evidence type	Supports / Refutes / Qualifies	Claim tested	Key finding	Context
PMID:25416781	Review of cited IDA source (database provenance)	Refutes (as β -oxidation evidence)	Does the GO:0033539 IDA reference actually assay GCDH β -oxidation?	Paper characterizes METTL20 methylation of ETF β ; GCDH only mentioned as an electron donor whose ETF transfer drops with methylation. No β -oxidation assay of GCDH.	Human, in (methyltran focus)
GO/QuickGO definition of GO:0033539; RHEA:13389	Ontology definition + reaction chemistry	Refutes	Does GCDH's reaction match the iterative β -oxidation definition?	GO:0033539 = iterative CoA-fatty-acid spiral ending in acetyl/propionyl-CoA; GCDH does a single decarboxylating dehydrogenation of a dicarboxylate.	Definitiona reaction da
PMID:15274622	Structural (crystal structures)	Refutes / qualifies	Nature of GCDH reaction	GCDH decarboxylates glutaryl-CoA to CO ₂ + crotonyl-CoA; distinct from β -oxidation.	Human GC protein
PMID:37198486	Cellular / pathway	Refutes (redirects to lysine catabolism)	Physiological process of GCDH	GCDH is the crotonyl-CoA-producing enzyme of lysine catabolism in human cells.	Human glioblastom
PMID:17176108	Direct assay (kinetics)	Qualifies	Reaction/ kinetics of GCDH	Oxidative decarboxylation of glutaryl-CoA; crotonyl-CoA release is rate-limiting.	Purified en

Citation	Evidence type	Supports / Refutes / Qualifies	Claim tested	Key finding	Context
PMID:21974953	Direct assay (substrate specificity)	Qualifies / partially supports (MF)	Is GCDH mechanistically an ACAD with fatty acyl capacity?	Catalytic properties very similar to short/medium- chain ACAD except added decarboxylation.	Rat GCDH, <i>in vitro</i>
PMID:11024031	Direct assay (mutagenesis + kinetics)	Partially supports (MF only)	Can human GCDH dehydrogenate a fatty acyl- CoA?	Hexanoyl-CoA (C6) is a measurable non- decarboxylated alternative substrate.	Human GCDH, <i>in vitro</i>
QuickGO withFrom provenance (Q92947)	Computational/ database provenance	Refutes (circularity)	Is the GO:0033539 IBA independent?	IBA withFrom = ['PTN005130278','Q92947'] — anchored on GCDH's own (misattributed) IDA.	GO_Central phylogenetic annotation
GO annotation distribution (GCDH, IVD, ACADM/S/VL/L)	Computational/ database	Refutes (paralog leakage)	Is GO:0033539 specific to true β -ox enzymes?	Term over-propagated to amino-acid-catabolism ACADs (GCDH, IVD) alongside true β -ox ACADs.	Cross-gene comparison

How the literature bears on the findings: The two crystallographic/kinetic papers ([PMID:15274622](#), [PMID:17176108](#)) and the human cellular study ([PMID:37198486](#)) collectively pin GCDH's physiological role to a single decarboxylating dehydrogenation within lysine catabolism, challenging the β -oxidation process claim. The two substrate-specificity/mutagenesis papers ([PMID:21974953](#), [PMID:11024031](#)) support only a molecular-level acyl-CoA-dehydrogenase capacity, not process participation. The cited IDA source ([PMID:25416781](#)) is the linchpin misattribution that, together with the circular IBA and the IEA transfer, generated the disputed annotations.

GO Curation Implications

Lead (requires curator verification):

GO term	Aspect	Current evidence on Q92947	Recommended action
GO:0033539 — fatty acid beta-oxidation using acyl-CoA dehydrogenase	BP	IDA (P 25416781 , misattributed) + IBA (circular, withFrom Q92947)	Remove as a GCDH process annotation. The IDA reference does not support β -oxidation; the IBA re-uses that IDA.
GO:0019395 — fatty acid oxidation	BP	IEA (Ensembl ortholog transfer from rat)	Remove / do not assert. Ortholog-transfer artifact; absent from the genuine FA-oxidation enzymes.
GO:0004361 — glutaryl-CoA dehydrogenase activity	MF	IDA + IBA on node PTN005130278	Retain — this is GCDH's core molecular function.
GO:0019477 / GO:0006554 — L-lysine catabolic process (glutaryl-CoA / lysine degradation)	BP	Supported by cellular evidence (P 37198486) and enzyme role	Retain / strengthen — this is GCDH's core biological process.
Generic acyl-CoA dehydrogenase activity (e.g., GO:0003995)	MF	In vitro hexanoyl-CoA activity (P 11024031 , P 21974953)	Optional, non-core. Could be captured at MF level with a clear "in vitro / promiscuous" note; do not upgrade to a fatty-acid β -oxidation BP term.

Summary of the curation call: The evidence supports **retaining GCDH's molecular function (glutaryl-CoA dehydrogenase) and its lysine/tryptophan-catabolism process** and **removing the two fatty-acid-oxidation biological-process terms (GO:0033539, GO:0019395)**. If a curator wishes to preserve the demonstrated in vitro fatty acyl-CoA activity, it belongs as a

non-core MF-level note, never as a β -oxidation process annotation. Note that this recommendation targets the *biological-process* claim specifically; it does not dispute GCDH's identity as an ACAD-fold enzyme.

Mechanistic Scope

The immediate molecular function tested is **flavin-dependent oxidative decarboxylation of glutaryl-CoA** (a dicarboxylic CoA thioester) to crotonyl-CoA + CO₂, with **ETF as the electron acceptor** — an ACAD-type dehydrogenation *plus* a decarboxylation. This is a discrete, single catalytic event.

Separating direct activity from downstream/inferred effects:

- **Direct gene-product activity:** glutaryl-CoA → crotonyl-CoA + CO₂ (GO:0004361); low-level in vitro dehydrogenation of straight-chain acyl-CoAs (hexanoyl-CoA).
- **Downstream/convergent metabolite:** crotonyl-CoA is *also* a β -oxidation intermediate, but GCDH producing it from glutaryl-CoA does not mean GCDH runs β -oxidation. Sharing a product is metabolic convergence, not process participation.
- **Loss-of-function phenotype (not evidence for β -oxidation):** GCDH deficiency causes glutaric aciduria type I (accumulation of glutaric and 3-hydroxyglutaric acid), a *lysine-catabolism* disorder — again pointing to amino-acid catabolism, not fatty-acid oxidation.

Conflicts and Alternatives

- **Paralog / family leakage:** GO:0033539 is shared across the ACAD family including amino-acid-catabolism members (GCDH, IVD). The term legitimately belongs to ACADM/ACADS/ACADVL/ACADL (which have independent IDA/IMP/ISS support) and appears to have leaked onto GCDH via superfamily membership and co-mention.
- **Circular provenance:** The IBA is anchored on GCDH's own misattributed IDA — a self-referential loop rather than independent phylogenetic support.
- **Text-mining misattribution:** The IDA source

(P 25416781)

is an unrelated methyltransferase paper; the same paper was bulk-attached to ACADM/P11310, consistent with an automated text-derived assignment.

- **Species differences:** The clearest substrate-specificity comparison

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P 21974953
)

is *rat* GCDH; the human hexanoyl-CoA data

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P 11024031
)

confirm the capacity is conserved, but neither addresses *in vivo* flux.

- **Legitimate counter-nuance (acknowledged):** ChEBI places glutaryl-CoA(5-) under omega-carboxy-(fatty acyl)-CoA(5-), and GCDH is mechanistically an ACAD — so the enzyme is not chemically excluded from acting on fatty acyl-CoAs. This nuance justifies an MF-level acyl-CoA-dehydrogenase note but does not rescue the β -oxidation *process* terms.

Limitations and Knowledge Gaps

1. Full-text of

P 25416781

assays: Access was at abstract/discussion level. The conclusion that it contains no GCDH β -oxidation assay is well supported by the abstract's framing (METTL20/ETF β), but a curator should confirm against the full Methods/Results that GCDH was used only as an electron donor. *Why it matters:* it is the sole experimental anchor for GO:0033539. *Resolution:* direct full-text inspection.

- ### 2. Quantitative relative activity:
- We know hexanoyl-CoA is an alternative substrate but did not extract the exact *k_{cat}/K_m* ratio versus glutaryl-CoA. *Why it matters:* quantifies how minor the fatty-acyl activity is. *Resolution:* recover kinetic tables from

P 11024031 /

P 21974953 .

- ### 3. No *in vivo* fatty-acid-oxidation flux data for GCDH:
- No study demonstrates GCDH contributing to cellular fatty-acid β -oxidation flux. *Why it matters:* absence of any *in vivo* evidence is the core reason to reject the BP terms. *Resolution:* isotope-tracing or GCDH-knockout FA-oxidation flux assays.

4. **IVD/other family annotations:** The over-propagation pattern was assessed at the annotation-listing level; a full audit of each family member's GO:0033539 provenance would strengthen the "leakage" claim. *Resolution:* systematic QuickGO provenance audit across the ACAD family.

Discriminating Tests

1. Full-text substrate audit of

P 25416781

— confirm GCDH is only an electron donor, not a β -oxidation substrate, in the cited IDA source. (Fastest, highest-yield.)

2. Kinetic ratio extraction — pull kcat/Km for glutaryl-CoA vs hexanoyl-CoA from

P 11024031

and the rat comparison in

P 21974953

to quantify how minor the fatty-acyl activity is.

3. Provenance audit across ACAD family — compare withFrom chains for GO:0033539 on GCDH, IVD, and the true β -ox ACADs to confirm the leakage/circularity pattern.

4. In vivo flux test (conceptual) — a GCDH-specific perturbation in a fatty-acid-oxidation flux assay would definitively separate physiological β -oxidation participation from lysine catabolism; predicted result: no FA-oxidation defect from GCDH loss (whereas MCAD/VLCAD loss impairs FA oxidation).

Proposed Follow-up Experiments / Actions (Curation Leads)

All items below are **leads requiring curator verification**.

Candidate action changes: - Remove GO:0033539 (fatty acid beta-oxidation using acyl-CoA dehydrogenase) IDA/IBA from Q92947 — IDA misattributed to

P 25416781

IBA circular. - **Remove / do-not-assert** GO:0019395 (fatty acid oxidation) IEA — ortholog-transfer artifact. - **Retain** GO:0004361 (glutaryl-CoA dehydrogenase activity) as core MF. - **Retain / strengthen** L-lysine catabolic process (GO:0019477 and related) as core BP. - **Optional non-core MF note:** generic acyl-CoA dehydrogenase activity, flagged as in vitro/promiscuous only.

Candidate **references** **with** **snippets** **to** **verify:** -

P 25416781

— "*METTL20-mediated methylation of ETF β in vitro reduced its ability to receive electrons from the medium chain acyl-CoA dehydrogenase and the glutaryl-CoA dehydrogenase*" → shows GCDH is only an electron donor; not a β -oxidation assay. -

P 15274622

— "*GCD catalyzes decarboxylation of glutaryl-CoA to produce CO(2) and crotonyl-CoA*" → confirms single decarboxylating dehydrogenation, not β -oxidation. -

P 37198486

— "*crotonyl-coenzyme A (crotonyl-CoA)-producing enzyme glutaryl-CoA dehydrogenase (GCDH)*" → amino-acid (lysine) catabolism context. -

P 21974953

— "*catalytic properties very similar to those of short-chain and medium-chain acyl-CoA dehydrogenase except its additional decarboxylation reaction*" → mechanistic ACAD, supports MF-level acyl-CoA capacity only. -

P 11024031

— "*alternative substrates, hexanoyl-CoA and glutaramyl-CoA, which are not decarboxylated*" → in vitro C6 fatty acyl-CoA activity, non-core.

Suggested curator questions: - Does the full text of

P 25416781

contain any GCDH β -oxidation assay? (Expected: no.) - Is GO:0033539 on IVD (via)

P 3597357

similarly a co-purification carry-over?

Suggested experiments: GCDH-perturbation fatty-acid-oxidation flux assay; quantitative kinetics comparing glutaryl-CoA and fatty acyl-CoA substrates.

Conclusion

The seed hypothesis raises legitimate nuances — GCDH is chemically an ACAD, glutaryl-CoA is a dicarboxylic acyl-CoA that ChEBI classifies under omega-carboxy-(fatty acyl)-CoA, and the enzyme can dehydrogenate a medium-chain fatty acyl-CoA in vitro — and those nuances correctly caution against dismissing all fatty-acyl capacity out of hand. But they do not sustain the disputed **process** annotations. The GO:0033539 IDA is misattributed to an unrelated METTL20/ETF β paper; its IBA is circularly anchored on that same bad IDA; GO:0019395 is an IEA ortholog-transfer artifact; and GCDH's actual chemistry is a single oxidative decarboxylation within lysine/tryptophan catabolism, not the iterative β -oxidation spiral GO:0033539 defines. **Recommendation: remove GO:0033539 and GO:0019395 as GCDH functions; retain glutaryl-CoA dehydrogenase activity and lysine catabolism as core; and, at most, represent any residual fatty-acyl activity as a non-core in vitro molecular function.**

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